EMPIRICAL CASE STUDY · 10-MINUTE TALK

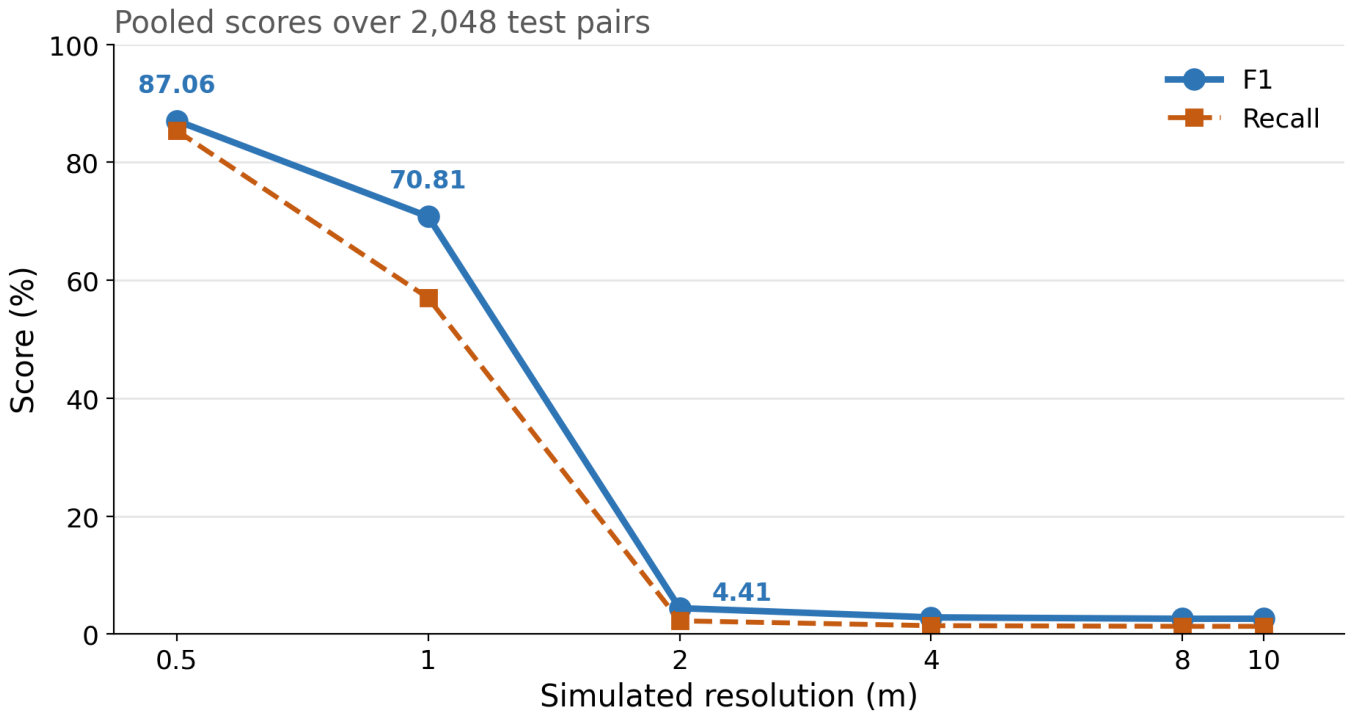
Resolution-Induced Performance Collapse of a CNN Change Detector

Simulated and real Sentinel-2 evidence from one frozen checkpoint

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WHAT WE MEASURED

One frozen MobileNetV2-based Siamese U-Net, trained once on 0.5 m LEVIR-CD and never retrained. We blur and resample 2,048 test pairs to 1, 2, 4, 8 and 10 m, then evaluate the same checkpoint. We also evaluate it zero-shot on the ten official OSCD test cities (3,077,936 pixels; 5.17% true change) against train-selected NDVI/NDBI rules and Otsu. The CNN weights are zero-shot; the joint-percentile preprocessing was selected on the training cities.



87.06% -> 4.41%
pooled F1, 0.5 m to 2 m
under blur + resampling

0.068%
pixels predicted changed at 10 m
91,006 of 134,217,728

924 / 935
changed images with zero
predicted change (98.82%)

Real OSCD (ten test cities, 3,077,936 pixels)

CNN joint F1 0.0007 · percentile 0.1176 · hardcoded 0.0782 · Otsu OA 0.3439 (F1 0.0953)

The test cities had been scored before the reported run. Descriptive scores, not an independent test.

CLAIM BOUNDARY

One checkpoint and one training run. Synthetic degradation, not a Sentinel-2 simulation.
OSCD confounds resolution with sensor, geography and season; no SCL in the benchmark.
No causal claim about the real Sentinel-2 gap. No priority or operational claim.
Spectral rules are not a validated fallback. The app is not validated end to end.

Saved evidence: degradation_results.csv; benchmarks/results/oscd_results_v2.json

No new test-city scoring was performed for this handout.

Paper, code, figures: TODO - insert verified repository URL and DOI.

Draft; citations and PDF build pending. Generated from saved results only.